

Coupling is required: cross-talk between policy and value pathways organises attention in a recurrent vision-transformer model of the primate visual system

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Abstract

Selective visual attention requires an animal to do two different things at once: decide, moment by moment, *what to do* on the basis of incoming sensory evidence, and estimate *how valuable* the current situation is, the quantity that supplies the learning signal for the first computation. In the primate brain these functions are carried by anatomically distinct but tightly interacting circuits — a sensory-grounded priority and commitment system in parietal, frontal and midbrain nodes (LIP, FEF, superior colliculus) and a value/prediction-error system in the cortico-basal-ganglia loop. Here we ask whether the *organisation* of these two computations, not merely their presence, is a determinant of whether attention-like behaviour can be learned at all. We study a recurrent vision-transformer agent trained purely from sparse reward to perform a spatially cued orientation-change-detection task modelled on primate covert-attention paradigms. We split the agent into two recurrent cross-attention pathways — a μ *pathway* that reads the live image and drives the policy (when to declare a change) and a q *pathway* that reads accumulated memory and drives the value estimate — and we vary a single structural property: whether the pathways *cross-talk* through a shared recurrent memory or run as isolated modules. The result is categorical. With cross-talk the agent becomes a calibrated psychophysical detector that reproduces the hallmark reliability-scaled cueing signature of primate spatial attention (**[acc / hit / FA / RT placeholders]**). Severing the coupling — giving each pathway a private memory — abolishes learning entirely: the agent never declares a change (**[hit=0]**). A second control that retains the coupling but routes the memory pathway to the policy collapses identically, establishing two jointly necessary conditions: the pathways must share a substrate, and the image-grounded pathway must command the decision. We trace the failure to credit assignment — a critic that evaluates a different state than the actor occupies cannot teach it — the same principle that binds dopaminergic value signals to cortico-striatal action learning. We further show the architecture generalises to larger stimulus arrays (a nine-stimulus set-size manipulation; **[set-size $\times$ validity result]**) and to environments with competing distractors (**[distractor result]**), and we use the trained model to generate falsifiable predictions — a representational and causal dissociation between the policy and value pathways — about what should be decodable from, and disrupted by perturbing, the corresponding neural circuits. The work illustrates how a reward-trained neural network, constrained to be interpretable and manipulable, can serve as a generative source of mechanistic hypotheses for systems neuroscience.

1 Introduction

Visual attention is the set of mechanisms by which a behaving animal commits its limited processing resources to the small fraction of the visual scene that is momentarily most relevant to its goals (Carrasco, 2011; Desimone et al., 1995; Reynolds and Heeger, 2009; Rust and Cohen, 2022). Decades of psychophysics and neurophysiology have established a robust catalogue of its behavioural and neural signatures: faster and more accurate responses to attended stimuli (Posner et al., 1980; Carrasco, 2011; Hawkins et al., 1990), the graded dependence of those benefits on the reliability and value of predictive cues (Mayo and Maunsell, 2016; Eckstein, 2011), and the modulation of single-unit responses and inter-areal synchrony throughout the visual hierarchy (Reynolds et al., 2000; Treue and Trujillo, 1999; Fries et al., 2001; Gregoriou et al., 2009). Yet a unified computational account of *why* attention is organised the way it is — of which structural features of the underlying circuitry are necessary for attentive behaviour to arise — remains incomplete (Mehrani and Tsotsos, 2023; Hassanin et al., 2024; Lindsay, 2020).

Attention as two coupled computations. A useful way to frame the problem is to recognise that an attending animal must perform two logically distinct computations concurrently. It must form a *policy* — a moment-by-moment decision about what to do, such as whether to report that a behaviourally relevant event has occurred — and it must form a *value estimate* — a prediction of how good its current situation is, which under any reward-driven account is precisely the quantity that supplies the teaching signal for the policy (Sutton, 2018; Dayan and Daw, 2008; Niv, 2009). The two computations draw on overlapping but distinguishable information. *When and where to act* is driven by bottom-up, image-grounded evidence accumulated toward a commitment threshold (Gold and Shadlen, 2007; Roitman and Shadlen, 2002; Ding and Gold, 2012); *how valuable the moment is* depends on accumulated context — which location was cued, how reliable the cue is, how much reward is at stake, how much time has elapsed (Platt and Glimcher, 1999; Sugrue et al., 2004; Stănişor et al., 2013). This division maps onto a deep distinction in the biology, between a sensory-grounded priority and commitment system and a value/prediction-error system, that we take as the central motivation for the present work.

The priority/commitment system. A network of topographically organised areas — the lateral intraparietal area (LIP), the frontal eye field (FEF), and the superior colliculus (SC) — maintains a “priority map” that combines bottom-up conspicuity with top-down task relevance and is read out to guide both overt orienting and covert attention (Bisley and Goldberg, 2010; Fecteau and Munoz, 2006; Thompson et al., 2005; Veale et al., 2017). The activity of these areas carries the signature of an evidence integrator: LIP firing ramps as the running time-integral of momentary sensory evidence toward a decision bound (Roitman and Shadlen, 2002; Huk and Shadlen, 2005), and FEF and SC convert that accumulation into a committed response (Ding and Gold, 2012; Stine et al., 2023). Causal manipulations place these areas upstream of attentive behaviour rather than merely correlated with it: sub-threshold microstimulation of FEF or SC *enhances* covert perceptual sensitivity at the corresponding location without an eye movement (Moore and Fallah, 2001; Moore and Armstrong, 2003; Cavanaugh et al., 2006; Müller et al., 2005), while their inactivation *impairs* covert selection (Wardak et al., 2006; Lovejoy and Krauzlis, 2010; Zénon and Krauzlis, 2012; Bollimunta et al., 2018). Critically, the contribution of the image-grounded signal is dissociable from the value it carries: in FEF the attentional effect is borne by visually responsive cells, with pure movement-related cells suppressed during covert attention (Thompson et al., 2005; Gregoriou et al., 2012), and in SC the deficit produced by inactivation survives intact cortical sensory gain (Zénon and Krauzlis, 2012).

The value/actor–critic system. Running in parallel, the cortico-basal-ganglia loop implements a reinforcement- learning actor–critic: action selection is associated with dorsal striatum (the “actor”), while value prediction and the reward-prediction-error teaching signal are associated with ventral striatum and midbrain dopamine (the “critic”) (Houk et al., 1995; Joel et al., 2002; O’Doherty et al., 2004; Maia, 2009; Collins and Frank, 2014). The cornerstone of this account is the close correspondence between phasic dopamine firing and the temporal-difference reward-prediction error (Schultz et al., 1997; Hollerman and Schultz, 1998; Montague et al., 1996), together with the dopamine dependence of cortico-striatal plasticity (Reynolds and Wickens, 2002; Morita and Kato, 2014). The two systems are not independent: the actor learns *only* through a three-factor plasticity rule in which the critic’s dopaminergic error is the gating third factor (Kuśmierz et al., 2017; Gerstner et al., 2018), so the value system and the action system are bound into a single closed learning loop. This furnishes a strong prior expectation — that policy and value computations cannot be severed without breaking learning — that our model makes precise and tests.

Existing computational models leave the organisational question open. Models of visual attention have captured many of its features one or a few at a time — normalisation-based gain control (Reynolds and Heeger, 2009), bottom-up saliency maps (Itti and Koch, 2001; Koch and Ullman, 1985), hierarchical selection (Tsotsos et al., 1995), guided visual search (Wolfe, 2021), and predictive/active-inference accounts (Feldman and Friston, 2010; Parr and Friston, 2017; Mirza et al., 2018). Few learn their attentional strategy from reward as animals do (Chelazzi et al., 2013; Anderson, 2016; Failing and Theeuwes, 2018; Mnih et al., 2014), and fewer still expose an interpretable, causally manipulable internal attention signal. Contemporary transformers, whose self-attention mechanism superficially resembles biological selection (Vaswani et al., 2017; Dosovitskiy et al., 2020; Khan et al., 2022), in practice emphasise low-level grouping rather than task-dependent, goal-driven selection (Mehriani and Tsotsos, 2023), and lack the recurrent internal state and capacity limits that are central to biological attention and working memory (Luck and Vogel, 1997; Awh et al., 2006; Panichello and Buschman, 2021). Recurrent neural networks supply the requisite dynamics (Hochreiter and Schmidhuber, 1997; Beck et al., 2024; Sussillo, 2014; Mante et al., 2013) but are not, by default, interpretable as attention maps. None of these traditions has asked the specific question we pose here: holding the learning signal and the components fixed, does the *wiring* between the policy and value computations determine whether attentive behaviour is acquired?

This work. We developed a recurrent vision transformer that integrates spatial working memory with self-attention into a “recurrent self-attention” mechanism and is trained end-to-end by reinforcement learning on sparse, animal-like reward (Springenberg et al., 2024; Bellemare et al., 2017). Building on a single-stream version of this model that already reproduces hallmark behavioural and causal signatures of primate covert attention, we partition the agent into two specialised recurrent cross-attention pathways: a μ *pathway* that queries the live image and drives the actor (the decision of when to declare a change), and a q *pathway* that reads accumulated memory and drives the distributional critic (the value estimate). We then manipulate a single structural variable — inter-pathway coupling — in the spirit of a lesion experiment, comparing (i) a *shared-memory* design in which the pathways cross-talk through a common recurrent state, (ii) an *independent* design in which each pathway is given a private memory it never shares, and (iii) a *swapped-routing* design that keeps the coupling but lets the memory pathway command the policy. We report four central findings. First, coupling is required: only the shared-memory design learns the task, and the independent design fails to acquire it at all. Second, the requirement is asymmetric: the image-grounded pathway must drive the policy; routing the decision through the memory pathway collapses the agent identically. Third, the coupled architecture is general — it scales to a nine-stimulus set-size manipulation and

to an environment with competing distractors, reproducing the attentional signatures in each. Fourth, the trained model yields concrete, falsifiable predictions for systems neuroscience: a representational dissociation (change-evidence should be decodable from the policy pathway, value/context from the value pathway) and a causal dissociation (perturbing the value pathway should spare detection while degrading value-based modulation, whereas perturbing the policy pathway should abolish detection). We use the deliberately interpretable, manipulable structure of the model not as an end in itself but as a tool: a generative source of mechanistic hypotheses that the experimental study of LIP, FEF, SC and the basal ganglia can test.

2 The model

2.1 Overview

The agent is a recurrent vision transformer trained by reinforcement learning to emit a binary decision (*wait* or *declare change*) on each frame of a spatially cued orientation-change-detection task (Section 3). Three components are connected in a loop (Figure [fig:model](#)). (1) A patch front-end parses each frame into a set of spatial tokens. (2) A recurrent cross-attention encoder integrates the current tokens with two per-token recurrent memories, maintained by long short-term memory (LSTM) cells across frames (Hochreiter and Schmidhuber, 1997; Beck et al., 2024). (3) An actor-critic reinforcement-learning module reads the encoder output to select an action and estimate value. The defining feature of the architecture is that the self-attention is *recurrent*: attention at each frame is computed not only from the immediate image but from the contents of a capacity-limited spatial working memory, so that the cue, shown briefly and then removed, can be maintained and used to bias later processing (Awh et al., 2006; Panichello and Buschman, 2021; Knudsen, 2007).

The encoder is organised into two parallel pathways. The μ *pathway* is grounded in the live image and feeds the actor (policy); the q *pathway* is grounded in accumulated memory and feeds the critic (value). The remainder of this section gives the precise form of the two pathways and the three structural configurations we compare.

2.2 Patch tokens and recurrent memories

Each 50×50 frame $\mathbf{O}^{(t)}$ is divided into a grid of non-overlapping patches, each embedded by a shared per-patch network and tagged with positional and temporal codes, yielding a set of tokens $\mathbf{X}^{(t)} = \{\mathbf{x}_i^{(t)}\}_{i=1}^N$ with $\mathbf{x}_i^{(t)} \in \mathbb{R}^d$. Two recurrent memories are carried as per-token hidden states (one slot per patch location), $\mathbf{H}_1^{(t)}, \mathbf{H}_2^{(t)} \in \mathbb{R}^{N \times d}$, updated frame to frame by independent LSTM cells. $\mathbf{H}_1$ is a fast sensory trace written from the raw image; $\mathbf{H}_2$ is a deep trace written from the policy pathway’s own output (below). The two memories play asymmetric roles — $\mathbf{H}_1$ keeps a recent record of the visual input, $\mathbf{H}_2$ accumulates the task-relevant state on which both the decision and its valuation depend.

2.3 The two cross-attention pathways

Each pathway is a single-head, pre-norm cross-attention block in which the *queries* are supplied by the current image tokens and the *keys and values* are supplied by memory. Writing $\text{Attn}(Q; K) = \text{softmax}(QK^\top/\sqrt{d})V$ with V tied to K , a residual R , and a position-wise feed-forward block FFN, a block computes

$$\mathbf{Z} = R + \text{Attn}(\mathbf{X}; \text{KV}), \quad \mathbf{Z} \leftarrow \mathbf{Z} + \text{FFN}(\mathbf{Z}). \quad (1)$$

The two pathways differ in their key/value bank and residual:

$$(\mu \text{ pathway, } \rightarrow \text{ actor}): \quad \mathbf{Z}_\mu^{(t)} = \text{block}(\mathbf{X}^{(t)}; \text{KV} = [\mathbf{H}_1^{(t)} \parallel \mathbf{H}_2^{(t)}], \text{R} = \mathbf{X}^{(t)}), \quad (2)$$

$$(q \text{ pathway, } \rightarrow \text{ critic}): \quad \mathbf{Z}_q^{(t)} = \text{block}(\mathbf{X}^{(t)}; \text{KV} = \mathbf{H}_2^{(t)}, \text{R} = \mathbf{H}_2^{(t)}). \quad (3)$$

The μ pathway queries *both* memories (concatenated along the token axis, each source tagged), and carries the raw image on its residual: it is the grounded, image-reading stream, and its decision can therefore reflect live visual evidence. The q pathway queries only the deep memory $\mathbf{H}_2$ and carries $\mathbf{H}_2$ on its residual: the image can only *re-weight* the stored state, a memory-grounded bottleneck appropriate to a contextual value estimate. The memories are then advanced,

$$\mathbf{H}_1^{(t+1)} = \text{LSTM}_1(\mathbf{X}^{(t)}, \mathbf{H}_1^{(t)}), \quad \mathbf{H}_2^{(t+1)} = \text{LSTM}_2(\mathbf{Z}_\mu^{(t)}, \mathbf{H}_2^{(t)}), \quad (4)$$

so that $\mathbf{H}_1$ is written from the raw image and $\mathbf{H}_2$ is written from the *policy pathway's* output $\mathbf{Z}_\mu$. Finally the heads read the two streams with a *split readout*: the actor reads $\mathbf{Z}_\mu$ and the (distributional) critic reads $\mathbf{Z}_q$,

$$\pi^{(t)} = \text{Actor}(\mathbf{Z}_\mu^{(t)}), \quad Q^{(t)} = \text{Critic}(\mathbf{Z}_q^{(t)}), \quad (5)$$

with the scalar value $V^{(t)}$ derived from the critic's quantile distribution under the current policy (Section 9).

2.4 The locus of coupling

The coupling between policy and value lives in $\mathbf{H}_2$. The actor's pathway *writes* $\mathbf{H}_2$, through $\mathbf{Z}_\mu$ in Equation (4), and *reads* it back, inside the concatenated bank of Equation (2); the critic's pathway *reads* the same $\mathbf{H}_2$ in Equation (3). The two heads are separated, but they sit on a shared, jointly maintained recurrent memory. This single shared register — a common “workspace” that the evaluative and the decision systems both read from and the decision system writes to — is the structural variable we manipulate.

2.5 Three pathway configurations

Holding the front-end, the heads, the parameter budget and the entire training recipe fixed, we compare three wirings (Figure [fig:configs](#)).

Configuration 1 — shared memory (cross-talk). Exactly Equations (2)–(5): two pathways coupled through the shared $\mathbf{H}_2$, with the grounded μ pathway driving the actor.

Configuration 2 — independent (no cross-talk). The same two pathway *types*, but each is given its own private memory, updated only by its own output:

$$\mathbf{Z}_\mu^{(t)} = \text{block}(\mathbf{X}^{(t)}; \text{KV} = \mathbf{H}_\mu^{(t)}, \text{R} = \mathbf{X}^{(t)}), \quad \mathbf{H}_\mu^{(t+1)} = \text{LSTM}_\mu(\mathbf{Z}_\mu^{(t)}, \mathbf{H}_\mu^{(t)}), \quad (6)$$

$$\mathbf{Z}_q^{(t)} = \text{block}(\mathbf{X}^{(t)}; \text{KV} = \mathbf{H}_q^{(t)}, \text{R} = \mathbf{H}_q^{(t)}), \quad \mathbf{H}_q^{(t+1)} = \text{LSTM}_q(\mathbf{Z}_q^{(t)}, \mathbf{H}_q^{(t)}). \quad (7)$$

The modules never touch each other's state; the only thing they share is the raw input image. This is the minimal edit that removes the cross-talk while preserving the split, the pathway types, the parameter count and the training recipe.

Configuration 3 — swapped routing. Architecturally identical to Configuration 1, but the readout is reversed: the q (memory) pathway drives the actor and the μ (image) pathway drives the critic. This isolates whether it matters *which* pathway commands the policy, independent of the presence of coupling.

The three configurations differ only in inter-pathway coupling and readout routing. The comparison therefore isolates the causal contribution of organisation, in the same spirit as a reversible inactivation experiment (Wardak et al., 2006; Lovejoy and Krauzlis, 2010).

3 Test environments

All experiments use spatially cued orientation-change-detection tasks built to reproduce the essential structure of the paradigms used to study covert attention in non-human primates (Posner et al., 1980; Cohen and Maunsell, 2009; Herman and Krauzlis, 2017; Srinath et al., 2021). The agent views a sequence of 50×50 frames and, on each frame, chooses to *wait* or to *declare a change*. The design isolates internal attentional state from sensory processing by separating the cue in time from the change it predicts (Posner et al., 1980; Müller and Findlay, 1987), and it makes attentional engagement parametric through cue *reliability* (validity) (Mayo and Maunsell, 2016).

3.1 Base task: cued change detection with value

Each trial unfolds over a fixed number of frames. The trial opens with a blank frame; a spatial cue is then presented briefly and removed, after which the display is blank again before the stimulus array appears; the array of oriented “Gabor”-like patches is then shown for the remainder of the trial. On half of trials, at a change time that is *jittered across frames*, the orientation of one patch steps by a magnitude Δ and persists; on the other half no change occurs. Frame-to-frame orientation noise is added throughout to titrate difficulty and to require genuine evidence integration rather than single-frame comparison.

The cue carries two pieces of information. Its *location* (here, a side — left or right of the array) indicates where a change is most likely to occur, and its *reliability* (validity) — the probability that a change, if one occurs, appears at the cued location — is drawn from $\{0.25, 0.50, 0.75, 1.00\}$ and is displayed by the completeness of a ring around the cue. The cue’s *colour* sets the reward magnitude available on that trial (a value manipulation, analogous to reward-biased attention paradigms (Stănişor et al., 2013; Baruni et al., 2015; Maunsell, 2004)). Reward is delivered for a correct decision: declaring at or after the change is a hit (rewarded by the cue-colour value); declaring before the change is a false alarm; waiting through a no-change trial is a correct rejection (rewarded); failing to declare a change is a miss. Because change onset is jittered and the response is event-locked, the agent cannot exploit a fixed temporal strategy and must instead maintain the cue and monitor for the change.

This base task is the setting for the central architectural comparison (Section 4). It combines a *validity* manipulation (graded cue reliability) with a *value* manipulation (cue colour), so that the trained agent can be probed both for the classic reliability-scaled cueing benefit and for value-directed responding.

3.2 Set-size variant: nine stimuli

To ask how attentional benefit scales with the number of competing locations — a question with a long history in the human and animal literatures (Palmer, 1995; Mazzyar et al., 2012; Eckstein, 2011) — we constructed a set-size variant with nine stimuli arranged in a 3×3 grid. A location cue marks one of the nine cells, and its ring completeness again signals reliability, now drawn from five levels $\{0, 0.25, 0.50, 0.75, 1.00\}$, where a proportion of 0 is an uninformative cue (uniform over all nine locations) and a proportion of 1 is a perfectly valid cue. Reward is uniform

Table 1. Behavioural outcomes of the three pathway configurations on the base task (500 episodes each). The shared-memory (cross-talk) agent is a calibrated detector; both the independent and swapped-routing agents collapse to never declaring a change. Numbers are the trained checkpoints reported in the cross-talk analysis; final-figure values to be regenerated.

Configuration	Accuracy	Hit rate	FA rate	RT (frames)	Phenotype
Shared memory (cross-talk)	0.868	0.794	0.067	1.95	active, calibrated detector
Independent (no cross-talk)	0.504	0.000	0.000	—	always-wait collapse
Swapped routing	0.420	0.000	0.000	—	always-wait collapse

across locations (removing the value manipulation) so that any reliability-graded effect reflects validity alone. With nine locations rather than four, knowing where to look should be worth more: the set-size variant tests whether the cue-reliability benefit, and the proportion-graded recruitment of attention, *grow* with set size — the set-size $\times$ validity interaction analysed in Section 5.

3.3 Distractor variant

The base task contains no competing events: when a change occurs it is always the task-relevant target. Natural attention, however, is defined by the need to select a target while *suppressing* salient but irrelevant events (Desimone et al., 1995; Sprague et al., 2018; Cosman et al., 2018). We therefore constructed a distractor variant in which, independently of the target, the uncued side may undergo its own orientation change with probability p_{dist} (with its own location, magnitude and timing). A change on the uncued side is a distractor that must be ignored: declaring in response to it is a false alarm that ends the trial without reward. Reward remains keyed to the target alone. The distractor variant thus turns the task into a selective-attention problem and lets us ask, of the split architecture, *where* in the two pathways an irrelevant event is filtered, and whether value modulates that filter (Section 6).

4 Coupling is required for learning

4.1 Cross-talk is the difference between a competent detector and a non-learner

We trained each of the three configurations of Section 2.5 on the base cued change-detection task (Section 3.1) with identical task, algorithm, hyperparameters and number of updates, and evaluated each on 500 fresh episodes under a stochastic policy. Outcomes were scored by signal-detection category — hit, miss, false alarm, correct rejection — and summarised by overall accuracy, hit rate on change trials, false-alarm rate on no-change trials, and reaction time on hits (Table 1, Figure [fig:behavior]).

The shared-memory agent is a genuine psychophysical detector: it catches roughly four of every five real changes, rarely fires on no-change trials ($\text{FAR} = 6.7\%$), and responds quickly — about two frames after change onset. The independent agent does not partially learn the task; across 500 episodes it *never once* declares a change. Its hit rate is exactly zero and its accuracy, 50.4%, is precisely the rate obtained by waiting out every trial and being correct only on the half that contain no change. The policy has retreated to the single safe action — always wait — that can never incur a false-alarm penalty, and it never discovers that declaring a change is ever worthwhile. The 50.4-percentage-point accuracy gap, and the 79.4-point hit-rate gap, between the coupled and uncoupled designs is not a matter of degree: it is the difference between a competent detector and a non-learner, produced by a single structural edit.

4.2 The collapse is specific to coupling, not to “having two streams”

The swapped-routing control rules out the trivial explanation that any split of policy and value into separate streams fails (or that any such split succeeds). Configuration 3 *keeps* the cross-talk and the shared memory and changes only which pathway issues the press decision; it collapses identically — zero hits, always wait. Two conditions are therefore jointly necessary. The pathways must be *coupled* through a shared memory (Configuration 2 fails without it), and the *grounded, image-reading pathway must drive the policy* (Configuration 3 fails when it does not). Only the configuration that satisfies both — the shared-memory agent with the μ pathway commanding the actor — learns. We return to why in Section 7.

4.3 The working agent reproduces the attentional cueing signature

A non-collapsed accuracy number is necessary but not sufficient evidence that the right computation has emerged. The discriminating test is whether the agent displays the behavioural fingerprint of covert spatial attention: a benefit for changes at the cued location that *scales with how reliable the cue is* (Posner et al., 1980; Carrasco, 2011; Mayo and Maunsell, 2016). Detailed characterisation of the shared-memory agent confirms this signature (Figure [fig:cueing]).

Reliability-scaled validity benefit. Splitting performance by cue reliability reveals a graded cueing effect: the advantage for a change at the cued location over an equivalent change at an uncued location *grows* as the cue’s reliability rises from 0.25 to 1.00 ([valid–invalid Δ by reliability]). At the lowest reliability, where the cue carries no predictive spatial information, there is no cueing effect; at the highest, a robust advantage emerges. The benefit is expressed primarily as a leftward shift of the psychometric function — a lower perceptual threshold for the attended location — mirroring the canonical sensitivity-enhancing signature of attention in primates (Treue and Trujillo, 1999; Reynolds et al., 2000; Cameron et al., 2002).

Value-directed responding. Because the cue colour sets the reward magnitude, the agent’s behaviour can also be read for value direction. Across the cue-colour reward structure the agent grades its responding by value ([value effect on criterion / response rate]), without abandoning low-value trials — the split between the policy and value pathways lets the value information shape behaviour without collapsing detection. The interaction between reliability and value, and the relative weight the agent places on each, is quantified in Section [sec:value-analysis].

Change-locked, fast decisions. The agent’s declarations are tightly locked to change onset (median reaction time ≈ 2 frames; Table 1), and attention allocated to the cued location is maintained through the blank interval after the cue and reactivated in anticipation of the change — the recurrent self-attention dynamics that a single-stream version of this model was previously shown to reproduce (Herman and Krauzlis, 2017; Ghose and Maunsell, 2002; Nobre and van Ede, 2018). The two collapsed variants, by contrast, have no behavioural signature to characterise: an agent that never presses produces a flat zero hit rate at every reliability and validity condition. The capacity to express spatial attention is downstream of the capacity to act at all, and only the coupled architecture clears that first bar.

5 Set size and the recruitment of validity-graded attention

A central prediction of any account in which attention allocates a limited resource is that the *benefit* of knowing where to look should grow with the number of competing locations (Palmer, 1995; Mazur et al., 2012; Eckstein, 2011). With four locations, a perfectly valid cue removes

a modest amount of spatial uncertainty; with nine, it removes much more, and an agent that exploits the cue should therefore recruit reliability-graded attention more strongly at the larger set size. We use the nine-stimulus variant (Section 3.2) to test this set-size $\times$ validity interaction in the coupled architecture.

Design and prediction. The nine-stimulus environment cues one of nine locations and signals its reliability through five proportion levels $\{0, 0.25, 0.50, 0.75, 1.00\}$, with an uninformative cue (proportion 0, uniform over all nine locations) as a within-task baseline and a uniform reward structure that removes the value manipulation, isolating validity. The architecture is the coupled cross-talk model of Section 2.5 applied unchanged. We make two quantitative predictions. First, the *behavioural* cueing benefit — the advantage for a change at the cued location, and its growth across proportion levels — should be *larger* at nine stimuli than at four. Second, and more diagnostically, the *attentional* recruitment should be more strongly graded by proportion: the attention the model allocates to the cued location at the moment of change should rise more steeply with cue reliability when there are nine competing locations than when there are four. In short, the prediction is that the nine-stimulus task uses proportion-graded attention *more* than the smaller-set-size tasks.

Status and analysis plan. We report this as a designed experiment and an open quantitative prediction rather than a completed result: at the time of writing the nine-stimulus run has not been trained to convergence with a logged, analysable model, and no per-reliability attention analysis exists ([**set-size $\times$ validity result — to be filled when the canonical sweep completes**]). The analysis is specified in advance. Using matched runs at set sizes $K \in \{2, 4, 9\}$ under one architecture and training recipe, we will (i) fit psychometric functions split by cue reliability and cued/uncued location and extract the reliability-graded validity benefit at each K ; and (ii) read the cross-attention weights at the change frame, compute the attention allocated to the cued versus uncued locations as a function of reliability, and test whether the reliability slope of cued attention increases with K . A positive interaction — steeper reliability-graded attention and a larger cueing benefit at $K = 9$ — would provide a parametric, within-model demonstration that the cue is recruited in proportion to the uncertainty it resolves, and a corresponding prediction for set-size manipulations in primate cueing experiments (Mayo and Maunsell, 2016; Eckstein, 2011).

6 Selective attention under competing distractors

The base task contains no competing events. To ask how the coupled architecture behaves when it must *select* a target against a salient irrelevant event — the defining demand of natural selective attention (Desimone et al., 1995; Sprague et al., 2018; Cosman et al., 2018) — we trained the same shared-memory cross-talk model (internal identifier `v11_part5`, 1,015,656 parameters) on the distractor variant of Section 3.3, in which the uncued side may undergo its own task-irrelevant change that the agent must ignore. The analysis reveals two selectivities riding on the single shared memory — one for the *decision*, one for *valuation* — and yields the clearest illustration in this work of the μ/q dissociation.

6.1 A value gate over a near-perfect detector

Aggregate performance understates the agent (accuracy ≈ 0.60 , hit ≈ 0.54 ; Figure [**fig:distr-behavior**]) because the aggregate averages over a deliberate refusal to engage on low-value trials. Split by cue colour — which sets reward magnitude — the picture is sharp: on high-value trials the agent is a near-perfect detector (hit ≈ 0.83 , false-alarm rate < 0.01 , $d' \approx 3.4$ – 3.9), whereas on the lowest-value trials it *abandons the task entirely*, declaring almost never (hit

≈ 0.00) and bailing at cue onset. The collapse is therefore not a perceptual failure but a value-driven *engagement* decision: when the reward on offer is too small, the agent declines to work for it. Tellingly, the same architecture trained on the distractor-*free* task does *not* abandon low-value trials — the bail is specific to the cost/benefit of selecting under competition. This is the model’s analogue of value-dependent task engagement, the kind of cost-of-control computation attributed to anterior cingulate and ventral striatal circuits (Maunsell, 2004; Stănişor et al., 2013).

6.2 Two selectivities for one decision: a causal dissociation

Clamping the attention of one pathway at the change frame and reading off behaviour and value separately produces a double dissociation (Figure [\[fig:dist-causal\]](#)). Biasing the μ (*policy*) *pathway* toward the cued location swings the decision across its full range (hit rate $0.01 \rightarrow 1.00$) while leaving the critic’s value estimate essentially unchanged; biasing the q (*value*) *pathway* moves the value estimate but never the decision. The two pathways share only the deep memory $\mathbf{H}_2$, yet one commands the decision and the other the valuation — a within-model demonstration that the split readout carries genuine functional content, not nominal labels.

6.3 The distractor is filtered in the decision pathway, not globally

Where is the irrelevant event suppressed? Not at the input: the distractor’s location is highly decodable from the fast sensory memory $\mathbf{H}_1$ (decoding accuracy ≈ 0.94), so the agent plainly *sees* it. It is suppressed specifically in the μ (*decision*) *pathway*, whose attention to the distractor is near zero (priority weight ≈ 0.004 versus ≈ 0.26 for the cued primary), while the q (*value*) *pathway* continues to read it. The suppression is causal: forcing the μ pathway’s attention onto the distractor converts the agent from a competent detector into one that false-alarms to the distractor (hit $1.00 \rightarrow 0.01$, false-alarm $0.00 \rightarrow 0.99$; d' inverts from $+5.5$ to -4.9). The distractor is thus “seen but not acted upon”: filtered in the pathway that issues the response rather than removed from the representation. This is a precise model counterpart of the superior-colliculus “dual effect” — unilateral SC inactivation simultaneously impairs responding to relevant events and *releases* responding to competing ones — in which the colliculus governs which signals inform the decision rather than which are sensed (Lovejoy and Krauzlis, 2010; Zénon and Krauzlis, 2012; Herman et al., 2020).

6.4 Value reaches into the decision filter, but as an engage/abandon switch

Finally, value does not merely set a post-perceptual response criterion: it reshapes the decision filter itself. On high-value trials the μ pathway’s allocation is sharp and the distractor strongly suppressed; on low-value trials the allocation is diffuse and leaky (cued/distractor weights $\approx 0.26/0.004$ for high value versus $\approx 0.03/0.11$ for low value). Crucially, this modulation is *binary* rather than graded over reward magnitude — the two engaged value levels behave alike, and the difference is whether the agent engages at all. Value, in this model, gates *whether* the priority filter is deployed, not how finely it is set, correcting a naive reading in which reward would simply scale a continuous criterion. The critic’s value estimates are themselves well calibrated and ordered by reward (calibration slope ≈ 1.0), consistent with a value system that prices the trial and a decision system that, gated by that price, either deploys selective attention or declines to.

6.5 Summary and predictions

The distractor variant shows the coupled architecture doing genuine selection: a near-perfect detector whose decisions are commanded by an image-grounded μ pathway that suppresses competitors at the level of action, gated by a q value pathway that prices the trial and decides

whether to engage. The mechanism makes two further predictions for the brain. First, a competing distractor should be *more* decodable from value/valuation populations than from the choice-driving populations that suppress it — the inverse of the usual assumption that suppression is global. Second, because suppression lives in the decision pathway, perturbing the priority node (SC/FEF) should release distractor responding while sparing the distractor’s sensory representation, whereas perturbing the value circuit should alter engagement and value-modulated filtering without abolishing detection (Zénon and Krauzlis, 2012; Herman et al., 2018; Gregoriou et al., 2012).

7 Why coupling is required

The failure mode points directly at its cause. The independent agent does not produce noisy or biased decisions — it produces *no* decisions, and it does so from the start of training onward. That is the signature of a broken learning signal, not a broken representation (Sutton, 2018; van Hasselt et al., 2018).

7.1 Credit assignment requires a shared state

Under an actor–critic objective, the policy improves by consulting the critic: the actor is pushed toward actions the critic values highly, evaluated *at the actor’s current state*. This only works if the critic’s value estimate refers to the same situation the actor is acting in. In the shared-memory agent that correspondence is enforced structurally. The actor acts from a representation built on $\mathbf{H}_2$ (Equation 2); the critic values a representation built on the *same* $\mathbf{H}_2$ (Equation 3); and the actor’s own output is what writes $\mathbf{H}_2$ (Equation 4). Policy and value are anchored to one evolving memory, so “pressing now looks good” is a statement about the state the actor actually occupies. Credit lands where it was earned, and the actor can discover that declaring a change at the moment of change is rewarded.

In the independent agent this correspondence is severed. The actor acts from its private memory $\mathbf{H}_\mu$; the critic values its private memory $\mathbf{H}_q$; the two drift apart because each is updated only by its own pathway. The critic learns an accurate value of states the actor never directly experiences, and the advantage signal handed to the actor is computed on the wrong representation. The result is a policy-improvement step that is systematically misattributed. With no coherent gradient toward the press action, the actor falls into the basin that is safe under *any* value estimate — never act, never be penalised for a false alarm — and stays there. The always-wait collapse is the rational endpoint of an agent that cannot tell when acting helps. This is a recognisable pathology in reinforcement learning: a representation mismatch between actor and critic, the failure mode that arises whenever a bootstrapped value function evaluates a different state distribution than the one the policy occupies (van Hasselt et al., 2018; Fujimoto et al., 2018). Architectural coupling is one way to guarantee, by construction, that the critic and actor are grounded in a common state.

7.2 The grounded pathway must drive the decision

The swapped-routing failure is the same lesson from the other side. There, value and policy *are* coupled through $\mathbf{H}_2$, but the actor is fed the q pathway, whose residual is memory rather than image (Equation 3): it carries almost no live visual content, only a re-weighting of stored state. The decision is then made from a representation that cannot see the change, so however clean the credit assignment, there is nothing in the actor’s input to assign credit *to*. A working agent needs both a faithful value signal and a grounded decision variable; cutting either one returns the same degenerate policy. This asymmetry has a direct counterpart in the biology: the attentional signal that drives behaviour is carried by the image-grounded, visually responsive populations of FEF and the SC, while the abstract/movement and value signals modulate but

do not substitute for it (Thompson et al., 2005; Gregoriou et al., 2012; Z  non and Krauzlis, 2012; Gold and Shadlen, 2007).

7.3 A shared workspace binding evaluation to decision

Functionally, the shared recurrent memory plays the role of a common “workspace” that both the evaluative and the decision systems read from and write to: value shapes what is maintained, and the maintained state shapes what is decided (Baars, 1988; Dehaene and Changeux, 2011; Miller and Cohen, 2001). The collapse under decoupling suggests that, in agents that must learn *when* to act from sparse, delayed reward, keeping these systems on a shared substrate is what makes the reward usable. The result is therefore about the *organisation* of computation, not its raw expressive power: the independent architecture has the parameters and the pathways to express the task, but the training signal it generates cannot drive them there. How attention is wired, not just how much machinery it has, decides whether the behaviour appears.

8 Discussion

The central result is architectural: for this class of recurrent attention agent, coupling between the policy and value pathways is a structural prerequisite for learning, on par with the choice of learning signal itself. A reasonable-looking modular design — two specialised recurrent streams, cleanly separated for policy and value — does not merely underperform when the streams are isolated; it fails to acquire the task entirely. The margin between the coupled and uncoupled designs is among the largest one can produce by a single structural edit, and it is bounded by two jointly necessary conditions: the streams must share a substrate, and the image-grounded stream must command the decision.

8.1 A mapping onto the primate attention and value systems

The model’s two pathways correspond, at the level of computational role, to the two systems introduced in Section 1, though the correspondence is a functional analogy rather than an anatomical isomorphism, and we state its limits plainly.

The μ pathway and the priority/commitment system. The μ pathway is image-grounded, drives the decision of *when* to declare a change, and writes the shared memory — a profile that aligns most naturally with the sensory-grounded priority-and-commitment system. The closest single-area analogue is the superior colliculus, which is causally necessary for covert selection (Lovejoy and Krauzlis, 2010; Z  non and Krauzlis, 2012; Bollimunta et al., 2018), whose sub-threshold stimulation enhances covert sensitivity without an eye movement (M  ller et al., 2005), and which acts as the threshold stage that converts parietal evidence accumulation into a committed response (Stine et al., 2023); its attentional contribution is moreover dissociable from cortical sensory gain (Z  non and Krauzlis, 2012) and feeds a subcortical decision loop through the basal ganglia (Herman et al., 2020). The frontal eye field supplies a second, cortical instance of the same role: it is a controller, not a read-out, of attention (Moore and Armstrong, 2003; Armstrong and Moore, 2007; Wardak et al., 2006; Gregoriou et al., 2009), and — crucially for our asymmetry result — its attentional signal is carried by visually responsive cells while pure movement cells are suppressed during covert attention (Thompson et al., 2005; Gregoriou et al., 2012). The lateral intraparietal area shows the integrator dynamics our recurrent memory instantiates (Roitman and Shadlen, 2002; Huk and Shadlen, 2005), but it straddles decision and value on a shared substrate (Platt and Glimcher, 1999; Sugrue et al., 2004) and is not causally indispensable in isolation (Katz et al., 2016; Jeurissen et al., 2022) — itself a useful caution against equating any single model module with one brain area.

The q pathway and the value/actor–critic system. The q pathway estimates value and, in the model, is more naturally identified with the value/critic machinery of the cortico-basal-ganglia loop — ventral striatum and dopamine (O’Doherty et al., 2004; Schultz et al., 1997; Joel et al., 2002) — than with any attention node. This is consistent with the biology, in which value signals for the oculomotor system live largely outside FEF/SC, in the basal ganglia and dopaminergic inputs (Hikosaka et al., 2006, 2014).

The coupling and the actor–critic teaching signal. The requirement that the two systems be coupled echoes the way dopaminergic value signals are bound to cortico-striatal action learning: the actor is updated only through a three-factor plasticity rule gated by the critic’s prediction error (Schultz et al., 1997; Reynolds and Wickens, 2002; Kuśmierz et al., 2017; Morita and Kato, 2014), so severing the value signal from the policy synapses breaks learning. We are explicit that the analogy is one of *computational principle*, not of mechanism: the brain’s coupling is a largely scalar, diffusely broadcast teaching signal flowing from critic to actor, whereas the model’s coupling is a shared recurrent working-memory register that the actor writes and both read. The model thus formalises one concrete way — a shared workspace — to satisfy a constraint that the basal-ganglia circuit satisfies by other means.

8.2 Predictions for systems neuroscience

Because the model is interpretable and manipulable, the dissociation it embodies becomes a set of falsifiable predictions at three levels.

1. *Representational.* Change-evidence (whether and where a change occurred) should be decodable preferentially from the policy (μ) pathway, and value/context (cued location, cue reliability, reward value) from the value (q) pathway. The neural prediction is that the decision/commitment nodes (SC, FEF visual cells, LIP choice signals) should carry the change-evidence variable, while value circuits carry the value variable (Thompson et al., 2005; Stine et al., 2023; Platt and Glimcher, 1999).
2. *Causal.* Perturbing the value pathway should spare detection while degrading value-based modulation, whereas perturbing the policy pathway should abolish detection — a double dissociation that maps onto value-circuit manipulation versus SC/FEF inactivation (Zénon and Krauzlis, 2012; Wardak et al., 2006; Herman et al., 2018).
3. *Behavioural-developmental (in the model).* Learning collapses when the value and policy systems are decoupled, predicting that interventions which functionally isolate value from action learning should impair the *acquisition* (not merely the expression) of attentive behaviour, consistent with the role of dopaminergic cortico-striatal coupling (Schultz et al., 1997; Collins and Frank, 2014).

We note honestly that the basal-ganglia actor–critic literature motivates the coupling and the existence of the split, but not the *asymmetry* (that the grounded pathway must drive the policy); that prediction rests instead on the evidence-accumulation literature, in which the sensory-grounded stage is the causally necessary one (Katz et al., 2016; Gold and Shadlen, 2007). The asymmetry is the model’s sharpest, most specific contribution.

8.3 Relation to the broader program and to other models

This finding sits alongside the broader thesis that primate-like attentional behaviour in these models emerges only under specific, jointly necessary constraints rather than from any single component: goal-driven reinforcement learning and memory-based gating of perception are each required, and here we add a third constraint of the same character — that policy and value

must share a recurrent substrate. To our knowledge no prior model of visual attention splits a *recurrent* encoder into policy and value streams, asks whether their coupling is required for learning, or maps the result onto basal-ganglia actor-critic and the priority nodes; attention-as-reinforcement-learning work has generally treated attention itself as the action (Mnih et al., 2014; Minut and Mahadevan, 2001), and the one machine-learning study of actor-critic representation sharing reports that, for *feed-forward* representations, separation can help — a contrast that sharpens our claim that for a *recurrent*, image-grounded agent coupling is instead required.

8.4 Limitations

The model is covert-attention only and omits overt orienting and the eye-movement system (Hoffman and Subramaniam, 1995; Deubel and Schneider, 1996); the patch-based bottleneck is adopted for interpretability rather than as a scalable theory of vision; the value pathway does not receive on-line value estimates directly but learns value-modulated behaviour from experience; and the neural mappings are functional analogies whose anatomical specificity we do not claim. Most importantly, the brain’s actor-critic coupling differs in substrate and direction from the model’s shared-memory coupling, so the biological results we cite are supporting analogies, not replications of the model’s exact manipulation. What the model offers is not a literal circuit but a generative, manipulable hypothesis-machine: a way to ask which organisational features of a policy/value system are necessary for attentive behaviour, and to read off concrete, testable predictions for the experiments that can adjudicate them. That a reward-trained network, constrained to be interpretable, can play this role is itself the broader point.

9 Methods

9.1 Patch front-end

Each $50 \times 50 \times 3$ frame is partitioned into a grid of square patches. For the main architectural comparison the image is divided into a 10×10 grid of 5×5 patches ($N = 100$ tokens); set-size and other variants use a grid matched to the stimulus array (Section 3). Each patch is embedded by a shared per-patch network into a $d = 128$ -dimensional token and augmented with learned positional and temporal codes. A variational-autoencoder front-end (Kingma and Welling, 2014) may optionally pre-encode the raw patches; in the configurations reported here a lightweight learned embedding is trained end-to-end with the agent. Information flows between spatial locations only through the attention mechanism; the recurrent memories are otherwise updated per token.

9.2 Recurrent cross-attention encoder

The encoder maintains two per-token recurrent memories $\mathbf{H}_1, \mathbf{H}_2 \in \mathbb{R}^{N \times d}$, advanced by independent LSTM cells (Hochreiter and Schmidhuber, 1997; Beck et al., 2024). On each frame the two pathways of Equations (2)–(3) are computed as single-head, pre-norm cross-attention blocks (Equation 1) whose queries are the current image tokens and whose key/value bank and residual are as specified per configuration (Section 2.5); a shared positional embedding is added to the memory keys, and the two memory sources entering the μ pathway’s key/value bank are source-tagged so that attention can distinguish $\mathbf{H}_1$ from $\mathbf{H}_2$. The memories are then updated by Equation (4). The three configurations differ only in the key/value bank, residual and readout routing; the parameter count is held fixed across them.

9.3 Actor–critic reinforcement learning

The actor and distributional critic are one-dimensional-convolutional heads over the token axis, reading the encoder streams as specified by Equation (5). The actor emits a wait/declare logit pair; the critic estimates a quantile distribution over returns (Dabney et al., 2018; Bellemare et al., 2017). Training uses an offline actor–critic objective: the actor loss is a maximum-a-posteriori policy-improvement term (Abdolmaleki et al., 2018; Springenberg et al., 2024) blended with a behaviour-cloning term, and the critic is trained with a quantile-Huber loss against a distributional Bellman target. A periodically hard-copied target network supplies both the critic’s bootstrap target and the policy-improvement reference. Episodes are stored in a prioritised replay buffer (Schaul et al., 2016), with priorities set by the per-episode quantile error, and a brief forced-wait warm-up trains the critic on full trajectories before the actor begins to act. Discounting uses γ ; learning rate, replay parameters and update counts are held fixed across the three configurations and reported in Table [tab:hparams].

9.4 Task parameters

Stimuli are hard-windowed oriented patches whose orientations are corrupted by frame-to-frame Gaussian noise to control difficulty; the change magnitude Δ is drawn per change trial and reduced over training by a performance-gated curriculum. Half of trials are change trials. The cue is presented on a single early frame and removed; the change, when present, occurs at a jittered time and persists. Reward is delivered for hits (scaled by the cue-colour value in the base task; uniform in the set-size variant) and for correct rejections. Exact timing, noise and reward values are listed in Table [tab:task].

9.5 Evaluation and analysis

Each trained configuration is evaluated on 500 fresh episodes under a stochastic policy and scored by signal-detection category (hit, miss, false alarm, correct rejection), from which we compute accuracy, hit and false-alarm rates, sensitivity d' and criterion c , and reaction time on hits (Green and Swets, 1966; Macmillan and Creelman, 2004). The cueing analysis splits hit rate by cue reliability and by cued/uncued change location; psychometric functions are fit with a four-parameter logistic and summarised by threshold, slope, guess and lapse parameters, with Bayesian credible intervals (Wichmann and Hill, 2001). The *pathway-dissociation* analysis (Section 8) trains linear decoders on each stream’s output at the change frame to predict change-evidence and value/context variables; the *causal-perturbation* analysis clamps the attention weights of a chosen pathway at the change frame and measures the effect on behaviour (for the μ pathway) and on the critic’s value estimate (for the q pathway). Attention maps are read directly from the cross-attention weights, one value per stimulus token, and averaged across trials within condition.

9.6 Reproducibility

The three configurations of the central comparison correspond to internal model identifiers `v11_part2` (shared memory / cross-talk), `v11_part4` (independent), and `v11_part3` (swapped routing); the distractor variant is `v11_part5` and the set-size variant is `setsize9`. Code, configurations and trained checkpoints are available at [repository URL].

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